

Ernest Choi

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Projects

General-Purpose Processor (ALU) | VHDL, FPGA, Quartus II, RTL

- Designed and implemented a modular Arithmetic and Logic Unit (ALU) in **VHDL** on an **FPGA**, integrating core components for hardware designs.
- Verified the design through **Quartus II** simulation, waveform analysis, and static timing verification, driving results out to 7-segment displays.
- Applied **RTL** design practice — timing analysis, modular integration, bitwise operations — to build and test digital logic at the register-transfer level.

eebot Mobile Robot System | C, Assembly, HCS12, PWM

- Developed embedded firmware in **Assembly** and **C** for a mobile robot built on the **HCS12** microcontroller, handling hardware verification and board-level integration in a Linux/Unix environment.
- Implemented **PWM** motor control with timer-driven scheduling that sustained 50ms sensor sampling intervals, driving the motor electronics from regulated voltage rails.

Particle Fluid Simulation | C++, OpenGL, GLSL, simulation github.com/ErChoi-com/Particle-Fluid-Simulation

- Developed a real-time fluid simulation in **C++** and **OpenGL**, writing **GLSL** shaders for particle rendering, physics-driven interaction, and GPU-accelerated parallel computation.
- Created a simple raycasting system, using projection math to simulate depth and perspective.
- Managed windowing, input, and render contexts with **GLFW** and **GLUT**, and trimmed graphics-pipeline memory allocation by 12% for smoother frames and better GPU utilization.

Experience

Electrical Team Member, Metropolitan Aerospace Rocket Society (MARS) – Toronto, ON September 2023 –

Present

- Designed, fabricated, and integrated **PCB** circuits using **Altium** and **KiCad** for custom STM32-based avionics flight computer with GPS telemetry, sensor integration, and pyrotechnic ignition systems
- Enhanced power efficiency by 23% through optimization of circuit configurations, component selection, and **power distribution** across regulated voltage rails
- Conducted board bring-up and debugging using **oscilloscopes** and **logic analyzers** to validate hardware functionality and ensure system reliability.
- Developed embedded firmware in **C/C++** and automated testing workflows with **Python** for microcontroller-based avionics.

Engineering Instructor, Obotz Robotics – Markham, Ontario

September 2022 – Present

- Coached students to place in the 75th percentile of robotics competitions with custom **Arduino** hardware.
- Taught digital logic concepts and embedded **C** programming, raising students' robotics problem-completion scores by 35%.
- Communicated students' technical progress to parents clearly.

Skills

Languages: JavaScript/Node.js, Java, Python, C++, C, SQL, HTML, CSS

Frameworks/Libraries: OpenGL, WebRTC, JavaFX, React, TypeScript, Flask, FastAPI, TensorFlow

Tools: Git, PostgreSQL, Ubuntu Linux, Docker, Kafka, Tableau, Power BI, FPGA

Platforms: GitHub, GitLab, Raspberry Pi, Arduino

Relevant Courses: Digital Systems, Microprocessor Systems, Object Oriented Eng Analysis and Design, Electronic Circuits I

Education

Toronto Metropolitan University – Bachelor of Engineering in Computer Engineering

2023–2027